

Lectures on Iteration Trees

Obrad Kasum

Gdańsk, Poland
April, 2025

1 April 22, 2025

Proposition 1.1. *Suppose*

1. M and N are inner models of ZFC,
2. $M \parallel (\kappa + 1) = N \parallel (\kappa + 1)$,
3. E is an M -extender and N -extender, with critical point κ .

Then $i_E^M \restriction (M \parallel (\kappa + 1)) = i_E^N \restriction (N \parallel (\kappa + 1))$ and

$$\text{Ult}(M, E) \parallel (i_E^M(\kappa) + 1) = \text{Ult}(N, E) \parallel (i_E^N(\kappa) + 1).$$

□

Notation 1.2. Let E be an extender. Then E is **nice** iff λ_E is inaccessible and $\text{Ult}(V, E) \parallel \lambda_E = V \parallel \lambda_E$. □

Notation 1.3. Let M be a transitive model of ZFC and let $\mathcal{T}$ be an iteration tree on M . Then $\mathcal{T}$ is **nice** iff for all α such that $\alpha + 1 < \text{len}(\mathcal{T})$,

- a. $E_\alpha^\mathcal{T}$ is nice in $\mathcal{M}_\alpha^\mathcal{T}$,
- b. for all $\xi < \alpha$, $\lambda_{E_\xi} < \lambda_{E_\alpha}$,
- c. $\text{pred}_\mathcal{T}(\alpha + 1)$ is the least $\xi \leq \alpha$ such that $\mathcal{P}(\kappa_{E_\alpha}) \cap \mathcal{M}_\xi^\mathcal{T} = \mathcal{P}(\kappa_{E_\alpha}) \cap \mathcal{M}_\alpha^\mathcal{T}$. □

Proposition 1.4. *Suppose M is a transitive model of ZFC and $\mathcal{T}$ is a nice iteration tree on M . Then for all $\alpha < \beta < \text{len}(\mathcal{T})$,*

- a. $\mathcal{M}_\alpha^\mathcal{T} \parallel \lambda(E_\alpha^\mathcal{T}) = \mathcal{M}_\beta^\mathcal{T} \parallel \lambda(E_\alpha^\mathcal{T})$,
- b. $\mathcal{M}_\alpha^\mathcal{T} \parallel (\lambda(E_\alpha^\mathcal{T}) + 1) \neq \mathcal{M}_\beta^\mathcal{T} \parallel (\lambda(E_\alpha^\mathcal{T}) + 1)$,
- c. $\text{pred}_\mathcal{T}(\alpha + 1)$ is the least $\xi \leq \alpha$ such that $\kappa(E_\alpha^\mathcal{T}) < \lambda(E_\xi^\mathcal{T})$. □

Notation 1.5. Suppose M is a transitive model of ZFC and $\mathcal{T}$ is a nice iteration tree on M . Then

- a. $\delta(\mathcal{T})$ is defined to be $\sup_{\alpha < \text{len}(\mathcal{T})} \lambda(E_\alpha^\mathcal{T})$,
- b. $\mathcal{M}(\mathcal{T})$ is defined to be $\bigcup_{\alpha < \text{len}(\mathcal{T})} \mathcal{M}_\alpha^\mathcal{T} \parallel \lambda(E_\alpha^\mathcal{T})$. □

Proposition 1.6. *Let δ be a limit ordinal. The following are equivalent.*

- a. δ is Woodin.
- b. For all $A \subseteq \delta$, there is $\kappa < \delta$ such that for all $\xi \in (\kappa, \delta)$, there is an elementary embedding $j : V \longrightarrow M$ such that
 - i. $\text{crit}(j) = \kappa$,
 - ii. $V_\xi \subseteq M$,
 - iii. $j(A \cap \kappa) \cap \xi = A \cap \xi$. □

Theorem 1.7. *Suppose*

- 1. $\mathcal{T}$ is a nice tree on V of limit length,
- 2. b and c are two distinct cofinal branches through $\mathcal{T}$,
- 3. $L(\mathcal{M}(\mathcal{T})) \models \text{ZFC}$.

Then $\delta(\mathcal{T})$ is a Woodin cardinal in $L(\mathcal{M}(\mathcal{T}))$.

Proof.

1° Let $\alpha := \sup(b \cap c)$. Note that $\alpha \in b \cap c$.

2° Let $\gamma_{-1} := \alpha$ and for $n < \omega$, let

- a. $\beta_n := \min(b - [0, \gamma_{n-1} + 1]) - 1$,
- b. $\gamma_n := \min(c - [0, \beta_n + 1])$.

3° Observe that for all $n < \omega$,

$$\beta_n + 1 \leq \text{pred}_\mathcal{T}(\beta_{n+1}) < \gamma_n + 1 \leq \text{pred}(\gamma_{n+1}) < \beta_{n+1} + 1$$

and that

$$\bigcup_{n < \omega} (\beta_n + 1) = \bigcup_{n < \omega} \gamma_{n+1} = \text{len}(\mathcal{T}).$$

4° **Claim.** For all $n < \omega$, $\kappa(E_{\gamma_n}^\mathcal{T}) < \lambda(E_{\beta_n}^\mathcal{T}) \leq \kappa(E_{\beta_{n+1}}^\mathcal{T}) < \lambda(E_{\gamma_n}^\mathcal{T}) \leq \kappa(E_{\gamma_{n+1}}^\mathcal{T}) < \lambda(E_{\beta_{n+1}}^\mathcal{T})$.

5° Let $A \subseteq \delta(\mathcal{T})$ be such that $A \in \mathcal{M}_b^\mathcal{T} \cap \mathcal{M}_c^\mathcal{T}$. We want to find $\kappa < \delta(\mathcal{T})$ such that for all $\xi \in (\kappa, \delta(\mathcal{T}))$, there is an elementary embedding

$$j : L(\mathcal{M}(\mathcal{T})) \longrightarrow N$$

satisfying that

- a. j is a definable class over $L(\mathcal{M}(\mathcal{T}))$,
- b. $\text{crit}(j) = \kappa$,
- c. $\mathcal{M}(\mathcal{T}) \parallel \xi \subseteq N$,
- d. $j(A \cap \kappa) \cap \xi = A \cap \xi$.

6° There are $m < \omega$, $B \in \mathcal{M}_{\beta_m}^{\mathcal{T}}$, and $C \in \mathcal{M}_{\gamma_m}^{\mathcal{T}}$ such that

$$i_{\beta_m, b}^{\mathcal{T}}(B) = i_{\gamma_m, c}^{\mathcal{T}}(C) = A.$$

We will show that we can take $\kappa := \kappa(E_{\beta_m}^{\mathcal{T}})$.

7° For $n < \omega$, let

$$F_n := E_{\beta_n}^{\mathcal{T}} \restriction (\kappa(E_{\gamma_n}^{\mathcal{T}}) + \omega) \text{ and } G_n := E_{\gamma_n}^{\mathcal{T}} \restriction (\kappa(E_{\beta_{n+1}}^{\mathcal{T}}) + \omega).$$

Note that for all $n < \omega$, $F_n, G_n \in \mathcal{M}(\mathcal{T})$.

8° Let $P_{m-1} := L(\mathcal{M}(\mathcal{T}))$ and for $n \in [m, \omega)$, let

- a. $N_n := \text{Ult}(P_{n-1}, F_n)$,
- b. $P_n := \text{Ult}(N_n, G_n)$.

9° **Claim.** For all $n \in [m, \omega)$,

- a. $N_n \parallel (\kappa(E_{\gamma_n}^{\mathcal{T}}) + \omega) = \mathcal{M}_{\beta_{n+1}}^{\mathcal{T}} \parallel (\kappa(E_{\gamma_n}^{\mathcal{T}}) + \omega)$,
- b. $P_n \parallel (\kappa(E_{\beta_{n+1}}^{\mathcal{T}}) + \omega) = \mathcal{M}_{\gamma_{n+1}}^{\mathcal{T}} \parallel (\kappa(E_{\beta_{n+1}}^{\mathcal{T}}) + \omega)$.

Proof. This follows from Proposition 1.1. □

10° Let $\xi \in (\kappa, \delta(\mathcal{T}))$ be arbitrary.

11° Let $n \geq m$ be such that $\lambda(E_{\beta_n}^{\mathcal{T}}) > \xi$ and let

$$j := i_{G_n}^{N_n} \cdots \circ i_{G_m}^{N_m} \circ i_{F_m}^{P_{m-1}} : L(\mathcal{M}(\mathcal{T})) \longrightarrow P_n.$$

Note that j satisfies the following:

- a. j is a definable class over $L(\mathcal{M}(\mathcal{T}))$;¹
- b. $\text{crit}(j) = \text{crit}(F_m) = \kappa_m = \kappa$,
- c. $\mathcal{M}(\mathcal{T}) \parallel \xi \subseteq \mathcal{M}(\mathcal{T}) \parallel \lambda(E_{\beta_n}^{\mathcal{T}}) = \mathcal{M}_{\gamma_{n+1}}^{\mathcal{T}} \parallel (\kappa(E_{\beta_{n+1}}^{\mathcal{T}}) + \omega) \subseteq P_n$.²

12° **Claim.** $j(A \cap \kappa) \cap \xi = A \cap \xi$.

Proof. This follows from Proposition 1.1. □

13° Claim 12° shows that j is as required. □

¹It is defined from extenders $F_m, G_m, \dots, F_n, G_n$.

²by Claim 9°

2 April 23 & 24, 2025

Proposition 2.1. *Suppose*

1. N_0 and N_1 are transitive models of ZFC,
2. M_0 and M_1 are countable transitive sets,
3. $j_0 : M_0 \longrightarrow N_0$ and $j_1 : M_1 \longrightarrow N_1$ are elementary,
4. E is a nice extender in M_1 , with critical point κ ,
5. $j_0 \upharpoonright (M_0 \parallel (\kappa + 2)) = j_1 \upharpoonright (M_1 \parallel (\kappa + 2))$,
6. $\text{Ult}(N_0, j_1(E))$ well-founded.

Then $\text{Ult}(M_0, E)$ is well-founded and there is a unique elementary embedding k such that the following diagram

$$\begin{array}{ccc}
 N_0 & \xrightarrow{i_{j_1(E)}^{N_0}} & \text{Ult}(N_0, F) \\
 j_0 \uparrow & & \uparrow k \\
 M_0 & \xrightarrow{i_E^{M_0}} & \text{Ult}(M_0, E) \\
 & & \uparrow j_1 \\
 & & M_1
 \end{array}$$

commutes and $k \upharpoonright (\text{Ult}(M, E) \parallel \lambda_E) = j_1 \upharpoonright (M_1 \parallel \lambda_E)$.

Proof. One can show that the only possible solution for k is given by

$$k([a, f]_E^{M_0}) := [j_1(a), j_0(f)]_{j_1(E)}^{N_0},$$

for $a \in [\lambda_E]^{<\omega}$ and $f : [\kappa]^{|\alpha|} \longrightarrow M_0$, with $f \in M_0$. One can then show that k has the desired properties. The well-foundedness of $\text{Ult}(M_0, E)$ follows from the existence of k . $\square$

Proposition 2.2. *Suppose*

1. M and N are transitive models of ZFC, where M is countable,
2. $j : M \longrightarrow N$ is elementary,
3. $\mathcal{T}$ is a nice tree on M .

Then there are unique $\mathcal{U}$ and $(j_\alpha : \alpha < \text{len}(\mathcal{T}))$ such that

- a. $\mathcal{U}$ is a nice tree on N ,
- b. the tree order underlying $\mathcal{U}$ is the same as that underlying $\mathcal{T}$,
- c. $j_0 = j$,

- d. for all $\alpha < \text{len}(\mathcal{T})$, $j_\alpha : \mathcal{M}_\alpha^\mathcal{T} \longrightarrow \mathcal{M}_\alpha^\mathcal{U}$ is elementary,
- e. for all $\alpha \mathcal{T} \beta$, $j_\beta \circ i_{\alpha,\beta}^\mathcal{T} = i_{\alpha,\beta}^\mathcal{U} \circ j_\alpha$,
- f. for all $\alpha \leq \beta < \text{len}(\mathcal{T})$, $j_\beta \upharpoonright (\mathcal{M}_\beta^\mathcal{T} \parallel \lambda_\alpha^\mathcal{T}) = j_\alpha \upharpoonright (\mathcal{M}_\alpha^\mathcal{T} \parallel \lambda_\alpha^\mathcal{T})$.

Proof. This follows from Proposition 2.1 by induction on $\text{len}(\mathcal{T})$. $\square$

Definition 2.3. In the context of the previous proposition, $\mathcal{U}$ is called the **copy** of $\mathcal{T}$ by j and $(j_\alpha : \alpha < \text{len}(\mathcal{T}))$ are called the **copy embeddings**. $\square$

Definition 2.4. Let $\mathcal{T}$ be a nice iteration tree on V of length ω . Then $\mathcal{T}$ is said to be **continuously ill-founded** iff there is a sequence of ordinals $(\alpha_n)_{n < \omega}$ such that for all $m, n < \omega$ satisfying $m \mathcal{T} n$, $i_{m,n}^\mathcal{T}(\alpha_m) > \alpha_n$. $\square$

Proposition 2.5. *There does not exist a continuously ill-founded tree on V .*

Proof.

- 1° Let us assume otherwise and let $\mathcal{U}^*$ be a nice iteration on V of length ω and let $(\beta_n^* : n < \omega)$ be a sequence of ordinals satisfying that for all $m, n < \omega$ with $m \mathcal{U}^* n$, $i_{m,n}^{\mathcal{U}^*}(\beta_m^*) > \beta_n^*$.
- 2° Let η be large enough with $\mathcal{U}^*, (\beta_n^*)_n \in V_\eta$.
- 3° Let $\theta > \eta$ be large enough and let $\mathcal{U}$ be the iteration on V_θ obtained with the same extender as $\mathcal{U}^*$. Note that for all $m, n < \omega$ with $m \mathcal{U} n$, $i_{m,n}^\mathcal{U}(\beta_m^*) > \beta_n^*$.
- 4° For $n < \omega$, let β_n be the $(\beta_n^*)^{\text{th}}$ regular cardinal above η in $\mathcal{M}_n^\mathcal{U}$. We have that for all $m, n < \omega$ with $m \mathcal{U} n$, $i_{m,n}^\mathcal{U}(\beta_m) > \beta_n$.
- 5° $\lambda_n < \eta < \beta_n$ for every $n < \omega$.
- 6° Let

$$k : (M, \in, \mathcal{T}, \delta, \alpha_n : n < \omega) \longrightarrow (V_\theta, \in, \mathcal{U}, \eta, \beta_n : n < \omega)$$

be elementary, where M is countable and transitive.

- 7° The sequence $(P_n, l_n : n < \omega)$ is constructed by recursion so that for all $n < \omega$,
 - a. P_n is a transitive model,
 - b. $l_n : M_n \parallel \alpha_n \longrightarrow P_n$ is elementary,
 - c. $l_n \in P_n$ and $|l_n|^{P_n} = \omega$,
 - d. for all $m < n$, $l_n \upharpoonright (M_n \parallel \lambda_m) = l_m \upharpoonright (M_m \parallel \lambda_m)$
 - e. if $n > 0$, then $P_n \in P_{n-1}$.

Once the construction is completed, we will have reached contradiction, as expressed by the last property.

8° For $n = 0$, let $P_0 := V_{\beta_0}$ and $l_0 := k \upharpoonright (M \parallel \alpha_0)$.

9° Assume $(P_i, l_i : i \leq n)$ has been defined. We want to define P_{n+1} and l_{n+1} .

10° Let m be the $\mathcal{T}$ -predecessor of $n + 1$. By Proposition 2.1, there is an elementary embedding l^* such that the diagram

$$\begin{array}{ccc} P_m & \longrightarrow & \text{Ult}(P_m, l_n(E_n)) \\ l_m \uparrow & & l^* \uparrow \\ M_m \parallel \alpha_m & \longrightarrow & \text{Ult}(M_m \parallel \alpha_m, E_n) \end{array} \quad \begin{array}{c} P_n \\ l_n \uparrow \\ M_n \parallel \alpha_n \end{array}$$

commutes and $l^* \upharpoonright \lambda_n = l_n \upharpoonright \lambda_n$, where the unnamed embeddings are canonical.

11° Since $M_{n+1} \parallel \lambda_n = M_n \parallel \lambda_n$ and λ_n is inaccessible in M_n , we have that

$$l^* \upharpoonright (M_{n+1} \parallel \lambda_n) = l_n \upharpoonright (M_n \parallel \lambda_n).$$

12° Let $\gamma := i_{m,n+1}^{\mathcal{T}}(\alpha_m)$, $F := l_n(E_n) \in P_n$, and $P^* := \text{Ult}(P_m, F)$. We have that

$$\text{Ult}(M_m \parallel \alpha_m, E_n) = \text{Ult}(M_m, E_n) \parallel \gamma = M_{n+1} \parallel \gamma$$

and the diagram

$$\begin{array}{ccc} P_m & \xrightarrow{i_F^{P_m}} & P^* \\ l_m \uparrow & & l^* \uparrow \\ M_m \parallel \alpha_m & \xrightarrow{i_{m,n+1}^{\mathcal{T}}} & M_{n+1} \parallel \gamma \end{array}$$

commutes

13° Let $\alpha^* := l^*(\alpha_{n+1})$. Since $\alpha_{n+1} < j_{m,n+1}(\alpha_m)$, we have that $\alpha^* < l^*(\gamma) = o(P^*)$.

14° The following holds:

- a. $P^* \parallel \alpha^*$ is a transitive model;
- b. $l^* \upharpoonright (M \parallel \alpha_{n+1}) : M \parallel \alpha_{n+1} \longrightarrow P^* \parallel \alpha^*$;
- d. for all $p < n + 1$, $l^* \upharpoonright (M_{n+1} \parallel \lambda_p) = l_p \upharpoonright (M_p \parallel \lambda_p)$.

15° **Claim.** There is l^{**} such that:

- b. $l^{**} : M \parallel \alpha_{n+1} \longrightarrow P^* \parallel \alpha^*$ is elementary;
- c. $l^{**} \in P^*$ and $|l^{**}|^{P^*} = \omega$;
- d. for all $p < n + 1$, $l^{**} \upharpoonright (M_{n+1} \parallel \lambda_p) = l_p \upharpoonright (M_p \parallel \lambda_p)$.

Proof.

- 1' Let $t := l^* \upharpoonright (M_n \parallel \lambda_n)$. We will ensure d. by ensuring $l^{**} \upharpoonright (M_n \parallel \lambda_n) = t$.
- 2' Note that $t \in P_n$ and $|t|^{P_n} = \omega$. The agreement between P_n and P^* ensures that $t \in P^*$ and $|t|^{P^*} = \omega$.
- 3' Let T be the set of all partial elementary embeddings

$$r : \{e_i : i < i_0\} \subseteq M_{n+1} \parallel \alpha_{n+1} \longrightarrow P^* \parallel \alpha^*,$$

where $i_0 < \omega$, such that for all $i < i_0$ for which $e_i \in \text{dom}(t)$, $t(e_i) = r(e_i)$. The order on T is the reverse inclusion. Observe that T is a tree.

- 4' Since $\alpha^*, t \in P^*$, we have that $T \in P^*$
- 5' The embedding l^* witness that T is ill-founded. Hence, it is ill-founded in P^* .
- 6' Let $(r_n)_{n < \omega} \in P^*$ be a branch through T and let $l^{**} := \bigcup_{n < \omega} r_n$. It is now easily seen that l^{**} is as required.

□

16° **Claim.** There are P_{n+1} and l_{n+1} such that:

- a. P_{n+1} is a transitive model;
- b. $l_{n+1} : M_{n+1} \parallel \alpha_{n+1} \longrightarrow P_{n+1}$ is elementary;
- c. $l_{n+1} \in P_{n+1}$ and $|l_{n+1}|^{P_{n+1}} = \omega$;
- d. for all $p < n + 1$, $l_{n+1} \upharpoonright (M_{n+1} \parallel \lambda_p) = l_p \upharpoonright (M_p \parallel \lambda_p)$;
- e. $P_{n+1} \in P_n$.

Proof.

- 1' Let $t := l^* \upharpoonright (M_n \parallel \lambda_n)$. We will ensure d. by ensuring $l_{n+1} \upharpoonright (M_n \parallel \lambda_n) = t$.
- 2' Let $\phi := l_n(\lambda_n)$. It is inaccessible both in P_n and P^* and $P^* \parallel (\phi + 1) = P_n \parallel (\phi + 1)$.
- 3' In P^* , there is $H \prec P^* \parallel \alpha^*$ such that $P^* \parallel \phi \cup \{l^{**}\} \subseteq H$ and $|H| = \phi$.
- 4' Let $\pi : H \rightarrow P_{n+1}$ be the transitive collapse and let

$$h : P_{n+1} \longrightarrow P^* \upharpoonright \alpha^*$$

be the anti-collapse.

- 5' Let $l_{n+1} := \pi(l^{**})$. We have that
 - $l_{n+1} : M_{n+1} \parallel \alpha_{n+1} \longrightarrow P_{n+1}$ is elementary,

- $l_{n+1} \in P_{n+1}$,
- $|l_{n+1}|^{P_{n+1}} = \omega$,
- $l_{n+1} \supseteq t$.³

6' Since P_{n+1} is coded by a subset of ϕ in P^* , it follows that $P_{n+1} \in P_n$.
Hence, P_{n+1} and l_{n+1} are as required.

□

17° Claim 16° concludes the proof.

□

³Since $t \in P^* \parallel \phi$, we have that $h(l_{n+1}) = l^{**} \supseteq t = h(t)$, which gives the conclusion.

3 April 25, 2025

Lemma 3.1. *Suppose*

1. M and N are inner models of ZFC,
2. F is a nice extender in N , with critical point κ ,
3. $M \parallel (\kappa + 1) = N \parallel (\kappa + 1)$,
4. $\text{Ult}(M, F)$ is well-founded,
5. $\eta \leq \kappa$ is an infinite cardinal,
6. both M and N are closed for η -sequences.

Then $\text{Ult}(M, F)$ is closed for η -sequences.

Proof. Let $([a_\xi, f_\xi]_F^M : \xi < \alpha)$ be a sequence of elements of $\text{Ult}(M, F)$. Note that

$$(a_\xi : \xi < \alpha) \in N \cap V_{\lambda_F} = \text{Ult}(N, F) \cap V_{\lambda_R} = \text{Ult}(M, F) \cap V_\lambda.$$

On the other hand, we have that

$$(i_F^M(f_\xi) : \xi < \eta) = i_F^M((f_\xi : \xi < \eta)) \restriction \eta \in \text{Ult}(M, F).$$

These two together yield

$$([a_\xi, f_\xi]_F^M : \xi < \eta) = (i_F^M(f_\xi)(a_\xi) : \xi < \eta) \in \text{Ult}(M, F).$$

□

Proposition 3.2. *Let $\mathcal{T}$ be a nice tree on V of length ω . Then for all $n < \omega$, $(\mathcal{M}_n^\mathcal{T})^\epsilon \subseteq \mathcal{M}_n^\mathcal{T}$.*

Proof. This is immediate from Lemma 3.1, by induction. □

Lemma 3.3. *Let $\mathcal{T}$ be a nice tree on V of length ω and let b be a cofinal branch through $\mathcal{T}$ such that $\mathcal{M}_b^\mathcal{T}$ is ill-founded. Then there are ordinals $(\alpha_n : n \in b)$ such that $i_{m,n}^\mathcal{T}(\alpha_m) > \alpha_n$ for every $m \in \mathcal{T} \restriction n \in b$.*

Proof. We consider the increasing enumeration $b = \{b_n : n < \omega\}$. Let

$$(i_{b_{n'_i}, b}(\beta'_i) : i < \omega)$$

be a decreasing sequence of ordinals in $\mathcal{M}_b$. Let $n_0 := 0$ and choose β_0 such that $j_{0, b_{n'_0}}(\beta_0) > \beta'_0$; define recursively $n_{i+1} := n_i + n'_i + 1$ and $\beta_{i+1} := j_{b_{n'_i}, b_{n_{i+1}}}(\beta'_i)$. In this way, we obtain an increasing sequence of naturals $(n_i : i < \omega)$, starting from 0, and a sequence of ordinals $(\beta_i : i < \omega)$ satisfying $j_{b_{n_i}, b_{n_k}}(\beta_i) > \beta_k$ for all $i < k$. Now, given an arbitrary $n < \omega$, there is a unique i such that $n_i \leq n < n_{i+1}$; we define $\alpha_{b_n} := \omega j_{n_i, n}(\beta_i) + n_{i+1} - n$. The sequence $(\alpha_m : m \in b)$ is as required. □

Theorem 3.4 (Martin, Steel). *Every nice tree on V of length ω has a well-founded cofinal branch. In other words, V is $(\omega + 1)$ -iterable.*

Proof. Assume otherwise. Let $\mathcal{T}$ be a nice tree on V of length ω without a well-founded cofinal branch. For $n < \omega$, let $\hat{n}$ denote the set of all cofinal branches through $\mathcal{T}$ that contain n ; the set $\hat{0}$ is exactly the set of all the cofinal branches through $\mathcal{T}$. (Some of these sets might be empty.) Notice that $m \triangleleft n$ implies that $\hat{n} \subseteq \hat{m}$. Given a branch $b \in \hat{0}$, the fact that $\mathcal{M}_b^{\mathcal{T}}$ is ill-founded implies, by Lemma 3.3, the existence of a sequence $(\alpha_n^b : n \in b)$ satisfying $i_{m,n}^{\mathcal{T}}(\alpha_m^b) > \alpha_n^b$, for all $m \mathcal{T} n \in b$.

Let R be the class of all pairs (m, f) , where $m < \omega$ and $f : \hat{m} \rightarrow \text{Ord}$, and let $<_R$ be a binary relation on it defined by

$$(m, f) <_R (n, g) : \iff n \mathcal{T} m \text{ and } (\forall b \in \hat{n})(f(b) < g(b)).$$

It is obvious that $<_R$ is a partial order on R ; let us verify that it is well-founded. Otherwise, there would be a sequence (n_i, f_i) ($i < \omega$) witnessing its ill-foundedness, so we would have a branch

$$b := \bigcup_{i < \omega} [0, n_i]_{\mathcal{T}} \in \bigcap_{i < \omega} \hat{n}_i$$

and consequently a sequence $f_0(b) > f_1(b) > f_2(b) > \dots$, which is a contradiction. Since $(R, <_R)$ is well-founded, it has a ranking function, say ρ .

By Proposition 3.2, $\hat{n}^{\mathcal{M}_k^{\mathcal{T}}} = \hat{n}$ and $(\hat{n} \rightarrow \text{Ord})^{\mathcal{M}_k^{\mathcal{T}}} = \hat{n} \rightarrow \text{Ord}$, for all $n, k < \omega$. In particular, $i_{0,n}^{\mathcal{T}}(R, <_R) = (R, <_R)$ and $i_{0,n}^{\mathcal{T}}(\rho) = \rho$. Let $\phi_n : \hat{n} \rightarrow \text{Ord}$ be defined by $\phi_n(b) := \alpha_n^b$. We have $(n, \phi_n) \in R$ and we can correctly define $\beta_n := \rho((n, \phi_n))$ ($n < \omega$). For $m \mathcal{T} n$, we want to check that $i_{m,n}^{\mathcal{T}}(\beta_m) > \beta_n$. In order to do so, it suffices to establish $i_{m,n}^{\mathcal{T}}((m, \phi_m)) >_R (n, \phi_n)$. Since $i_{m,n}^{\mathcal{T}}((m, \phi_m)) = (m, i_{m,n}^{\mathcal{T}}(\phi_m)) = (m, i_{m,n}^{\mathcal{T}} \circ \phi_m)$ and we already know that $m \mathcal{T} n$, it remains to verify that $(i_{m,n}^{\mathcal{T}} \circ \phi_m)(b) > \phi_n(b)$ for all $b \in \hat{n}$, which is just another way of saying $i_{m,n}^{\mathcal{T}}(\alpha_m^b) > \alpha_n^b$. Thus, the tree $\mathcal{T}$ is continuously ill-founded, which contradicts Proposition 2.5. $\square$

References

- [Kas21] Obrad Kasum. *Projective determinacy*. Master's Thesis, Université Paris Cité. See <https://obradkasum.github.io/notes/>. 2021.

Index

$\delta(\mathcal{T})$, 2

$\mathcal{M}(\mathcal{T})$, 2

continuously ill-founded, 5

copy, 5

copy embeddings, 5

nice, 1